IoT Sensor Failure Prediction: An AutoML-Based Predictive Maintenance System for Rotating Industrial Machinery

Supriya Balakrishna
Department of Computer Science and Engineering (AI & ML)
Alliance University
Bengaluru, Karnataka, India
bsupriyaBTECH23@ced.alliance.edu.in

Dr. Marimganti Srinivas
Associate Professor
Alliance University
Bengaluru, Karnataka, India
marimganti.srinivas@alliance.edu.in

***Abstract*—Predictive maintenance (PdM) plays a critical role in Industry 4.0 by enabling continuous health monitoring of industrial rotating machinery to mitigate unscheduled downtime, reduce maintenance overhead, and eliminate catastrophic structural failures. However, standard machine learning (ML) diagnostic workflows frequently rely on manual hyperparameter tuning, fail to generalize across dynamic torque load variations, and lack end-to-end integration with asynchronous Industrial Internet of Things (IIoT) telemetry pipelines. This paper presents an integrated, data-driven predictive maintenance architecture for five-class machinery fault diagnosis (**Normal**, **Inner Race Fault [BPFI]**, Outer Race Fault [BPFO]**, **Shaft Misalignment**, and **Rotor Unbalance**) using multi-channel vibration, motor current, and thermal telemetry. Grounded in the benchmark dataset published by Jung et al. and hosted in the Mendeley Data repository (comprising 45 matched multi-sensor recordings collected across 0 Nm, 2 Nm, and 4 Nm static torque loads), our system extracts 1-second non-overlapping statistical and spectral feature windows to form a standardized 52-dimensional feature schema. We evaluate eight candidate estimators alongside an automated hyperparameter search engine (FLAML with a 20-second time budget) under strict **recording-level data partitioning** to eliminate window-level data leakage. Standard Random Forest (200 decision trees) emerged as the top-performing model, achieving an overall held-out test accuracy of **87.50%**, a balanced accuracy of **80.00%**, and a macro-averaged F1-score of **77.14%** across 960 unseen test windows. Detailed error analysis demonstrates flawless fault isolation (100% F1-score) for BPFI, BPFO, and Misalignment defects. However, the evaluation uncovers a critical diagnostic failure mode: the Normal operating class suffered a **0.00 recall**, with all 120 Normal test windows misclassified as Unbalance due to severe dataset imbalance (only 3 total Normal recordings available). Furthermore, cross-load evaluation revealed performance degradation under high torque loads (macro-F1 dropping to 0.5000 at 4 Nm), while post-hoc probability calibration checks yielded an Expected Calibration Error (ECE) of 0.4090 and produced **0 selective-prediction coverage** (100 operator referral rate) under a 90% target validation accuracy policy. Finally, we demonstrate the operational microservice deployment of this pipeline utilizing asynchronous **MQTT** telemetry replay, a **FastAPI** prediction worker, an **SQLite/PostgreSQL** event store, and an interactive **Streamlit** operator dashboard.**

***Index Terms*—Predictive Maintenance, Industrial IoT, AutoML, Fault Classification, Rotating Machinery, Multisensor Data, MQTT, Random Forest.**

I. Introduction

Rotating machinery—including induction motors, drive shafts, centrifuges, gearboxes, and industrial pumps—forms the mechanical backbone of modern industrial manufacturing. Within these assemblies, rolling element bearings (REBs) and coupling shafts endure continuous cyclic mechanical loads, rotational friction, and dynamic operational stress. Over extended operation, these components inevitably succumb to surface fatigue, raceway spalling, structural misalignments, and rotor mass unbalance. Historically, industrial maintenance paradigms have oscillated between two conventional strategies[1],[2]: 1. *Reactive Maintenance ("Run-to-Failure"):* Equipment is operated until a functional breakdown occurs. While this strategy eliminates routine inspection costs, it introduces severe operational hazards, unpredictable production line stoppages, expensive secondary collateral damage to adjacent machinery, and prolonged recovery lead times. 2. *Preventive Maintenance (Time-Based Overhauls):* Maintenance routines are conducted at fixed calendar or operating-hour intervals regardless of actual equipment physical degradation. Although preventive schedules mitigate catastrophic failures, they frequently lead to premature component replacements, unnecessary labor expenses, and operational disruptions for healthy machinery. With the advent of Industry 4.0 and the Industrial Internet of Things (IIoT), **Predictive Maintenance (PdM) has emerged as a superior, condition-based alternative. By equipping industrial machinery with multi-modal sensor arrays—such as tri-axial Micro-Electro-Mechanical Systems (MEMS) accelerometers, high-frequency current transducers (CTs), and thermocouples—operators can capture continuous physical telemetry reflecting structural vibration, electrical power consumption, and thermal dissipation. Processing these multi-dimensional time series via machine learning algorithms allows industrial facilities to detect early structural wear, isolate localized fault modes, and schedule maintenance interventions before catastrophic failure occurs.

A. Research Problem and Motivation

Despite significant advancements in data-driven fault diagnosis, deploying automated predictive maintenance pipelines within operational industrial environments faces three primary technical obstacles: * **Manual Optimization Bottlenecks:** Traditional machine learning workflows (e.g., Support Vector Machines, Random Forests) and Deep Neural Networks (DNNs) depend heavily on manual feature extraction, signal filtering selection, and hyperparameter tuning. Designing optimal model architectures for dynamic sensory signals requires specialized domain expertise, creating an implementation barrier for industrial enterprises. * **Evaluation Data Leakage in Published Benchmarks:** A prevalent methodological flaw in academic literature is the application of random sample- or window-level data splitting across continuous time-series recordings. When 1-second sliding windows extracted from the same physical machine run are split randomly into training and testing sets, temporal auto-correlation enables the model to memorize physical signatures rather than learning generalizable fault dynamics. This data leakage yields artificially inflated accuracy metrics (frequently exceeding 99) that fail to hold when tested on unseen machinery runs. * **Operational Load Variations and Distribution Shifts:** Industrial machinery rarely operates under constant static conditions. Fluctuating rotational speeds and dynamic torque loads (e.g., 0 Nm, 2 Nm, 4 Nm) alter vibration signal energy distributions, shift fundamental fault characteristic frequencies, and change phase current draws. Consequently, diagnostic models trained on static single-load datasets suffer severe performance degradation when deployed across dynamic operational regimes.

Problem Statement: The primary objective of this work is *to create a predictive maintenance system using AutoML techniques that analyzes IoT sensor data to forecast equipment failures, minimize downtime, and improve operational reliability in industrial environments.* This study specifically implements and evaluates a five-class machine condition and fault classification framework, establishing the foundational diagnostic layer required for future time-to-failure forecasting.

B. Specific Contributions

Leakage-Free Recording-Level Evaluation Protocol:** We demonstrate that traditional random window splitting inflates diagnostic accuracy, and establish a strict recording-level data partitioning protocol across 45 matched multi-sensor runs to provide an honest assessment of model generalization. * **Unified Multi-Modal 52-Feature Schema:** We construct a robust signal processing schema synthesizing 4 vibration channels, 2 housing temperature channels, and 1 common phase current channel (‘ai0‘), explicitly addressing missing sensory channels in open industrial benchmarks. * **Rigorous AutoML &; Estimator Comparison:** We conduct a comparative evaluation of eight candidate estimators (including SVM, Extra Trees, XGBoost, LightGBM) alongside FLAML AutoML, establishing that standard Random Forest (200 trees) yields superior validation macro-F1 performance (0.8125) over FLAML’s independently selected LightGBM model (0.7810). * **Transparent Analysis of Diagnostic Vulnerabilities:** We explicitly document critical model failure modes, exposing a **0.00% recall on the Normal health state** caused by dataset imbalance, quantifying cross-load degradation at 4 Nm torque (macro-F1 of 0.5000), and proving that post-hoc calibration leads to **0* **Production-Ready Containerized Software Microservice:** We deliver a complete software implementation featuring historical MQTT telemetry replay (QoS 1 with deduplication), FastAPI prediction endpoints, persistent database event logging (SQLite/PostgreSQL with Alembic migrations), and an interactive Streamlit condition monitoring UI.

II. Related Work

A. Signal Processing and 2D Image Encoding Approaches

A dominant trend in machinery fault diagnosis is transforming 1D sensor signals into 2D spatial images to leverage pre-trained Convolutional Neural Networks (CNNs). Toma et al.[3] demonstrated that converting 1D stator current signals into 2D Gramian Angular Field (GAF) images allows a 2-layer CNN to achieve classification accuracies exceeding 99% across varying operating conditions on the UPB dataset. Similarly, Lee et al.[4] combined an Improved Fast Kurtogram (IFK) with Nonlinear Mode Decomposition (NMD) and GAF image transformations, proving that filtered time-frequency images enable 2D CNNs to maintain high diagnostic robustness under heavy Gaussian background noise. Xie et al.[5] evaluated Continuous Wavelet Transform (CWT) time-frequency representations paired with CNN feature extractors, showing that hybrid CNN-Random Forest classifiers maximize feature representation relative to standalone SVMs.

However, as highlighted by Fort et al.[6], 2D image transformation algorithms (GAF, CWT, Short-Time Fourier Transform) introduce massive computational overhead, memory expansion, and latency spikes during matrix transformations. This renders 2D image encoding ill-suited for low-power edge microcontrollers or real-time high-frequency streaming applications.

B. Edge Diagnostics and Low-Cost Sensing

To achieve cost-effective deployment, recent studies explore lightweight anomaly detection implemented directly on microcontrollers. Kolok et al.[7] engineered a low-cost predictive maintenance hardware node using an ESP32-C6 microcontroller paired with an MPU6050 MEMS accelerometer, utilizing an unsupervised Isolation Forest model to detect mechanical wear while keeping total hardware expenses under €30Ḱuo et al.[8] developed a sensor node utilizing lower-cost MEMS sensors (ADcmXL3021 and Hi229), employing Wavelet Packet Decomposition (WPD), Principal Component Analysis (PCA), and Long Short-Term Memory (LSTM) networks to achieve a 97% fault classification accuracy.

Despite their low cost, resource-constrained edge microcontrollers often restrict processing to basic time-domain statistical metrics (RMS, Peak-to-Peak). They frequently struggle

to maintain diagnostic accuracy when subjected to dynamic speed fluctuations and dynamic torque shifts.

C. Automated Machine Learning (AutoML) in Industrial IoT

AutoML frameworks (e.g., PyCaret, FLAML, AutoKeras) automate data preprocessing, algorithm selection, hyperparameter tuning, and pipeline packaging. Hadi et al.[9] applied PyCaret AutoML and AutoKeras AutoDNN models to the CWRU bearing dataset, reporting a 99.70% test accuracy using an AutoML-selected Random Forest. In a comprehensive systematic review, Marín Vásquez et al.[2] analyzed AI algorithms and software IoT platforms, identifying a critical gap in existing literature: while AutoML simplifies model creation, published works rarely bridge the gap between static model generation and operational industrial IoT platform integration (e.g., asynchronous MQTT message brokers, API microservices, and database persistence). Table I presents a comparative analysis of representative studies, highlighting their sensor modalities, methodologies, reported performance, and limitations.

III. Dataset and Methodology

A. Mendeley Data Repository Dataset Sources

This study utilizes the open benchmark dataset entitled *"Vibration, Acoustic, Temperature, and Motor Current Dataset of Rotating Machine Under Varying Operating Conditions for Fault Diagnosis"*, published by Jung et al.[10] and hosted in the Mendeley Data repository (DOI: ‘10.17632/ztmf3m7h5x.6‘, dataset version 5). The collection captures synchronized multi-sensor signals from a rotating machine testbed operating under three static torque load conditions (0 Nm, 2 Nm, and 4 Nm) at a constant rotational speed of 3010 RPM.

The dataset contains **45 matched multi-sensor recordings**: * **Normal (Healthy):** 3 Recordings (120 seconds each) * **Inner Race Fault (BPFI):** 9 Recordings (0.3 mm, 1.0 mm, 3.0 mm damage severity; 60 seconds each) * **Outer Race Fault (BPFO):** 9 Recordings (0.3 mm, 1.0 mm, 3.0 mm damage severity; 60 seconds each) * **Shaft Misalignment:** 9 Recordings (0.1 mm, 0.3 mm, 0.5 mm offset; 60 seconds each) * **Rotor Unbalance:** 15 Recordings (583 mg to 3318 mg unbalance mass on Disk 4; 60 seconds each) Data Hygiene & Channel Selection:** Data validation revealed that motor phase current channels ‘ai2‘ and ‘ai3‘ in the TDMS files are completely unpopulated across all 9 BPFO fault recordings. To guarantee a consistent schema across all 45 matched runs, ‘ai2‘ and ‘ai3‘ were removed, retaining **ai0** as the sole common motor current feature. Acoustic streams were also excluded due to incomplete class coverage across static load variations.

B. Signal Processing and 52-Dimensional Feature Extraction

Raw time-series telemetry streams are segmented into non-overlapping 1-second windows (25,600 samples per window at 25.6 kHz). For each window, statistical time-domain and frequency-domain spectral metrics are computed across all 7 common channels:4 Vibration Acceleration Channels Triaxial acceleration measured on Housing A and Housing B. Extracted features (8 per channel): Mean, RMS, Standard Deviation, Peak-to-Peak, Kurtosis, Crest Factor, Dominant Frequency, and Spectral Centroid. (32 features).2 Housing Temperature Channels : Thermocouple telemetry on Housing A and B. Extracted features (6 per channel): Mean, RMS, Standard Deviation, Peak-to-Peak, Kurtosis, and Crest Factor. (12 features).1 Motor Stator Current Channel Phase U current transducer telemetry. Extracted features (8 per channel): Mean, RMS, Standard Deviation, Peak-to-Peak, Kurtosis, Crest Factor, Dominant Frequency, and Spectral Centroid. (8 features).This yields a standardized 52-dimensional feature vector per 1-second window. All features are normalized using RobustScaler to handle transient mechanical impulses:

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$$

$$\text{Kurtosis} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^4}{\left[\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2\right]^2}$$

$$\text{Crest Factor} = \frac{\max(|x_i|)}{\text{RMS}}$$

$$\text{Spectral Centroid} = \frac{\sum_{k=1}^{K} f(k) \cdot |X(k)|}{\sum_{k=1}^{K} |X(k)|}$$

$$x_{\text{scaled}} = \frac{x - Q_2(x)}{Q_3(x) - Q_1(x)}$$

Target health states are ordinal-encoded: Normal: 0, BPFI: 1, BPFO: 2, Misalignment: 3, Unbalance: 4.

C. Leakage-Free Recording-Level Partitioning

To prevent window-level data leakage, complete recordings are assigned strictly to individual partitions:

Training Partition: 25 matched recordings ($\tilde{2}$,760 1-second windows; contains 1 Normal recording). Validation Partition: 10 matched recordings ($\tilde{1}$,080 1-second windows; contains 1 Normal recording). Held-Out Test Partition: 10 matched recordings (960 1-second windows; contains 1 Normal recording).

D. Candidate Estimator Search and AutoML Benchmarking

Eight candidate algorithms were evaluated alongside an automated hyperparameter search using FLAML (configured with a **20-second time budget** optimizing for macro-averaged F1-score):

1. Dummy Prior Classifier 2. Logistic Regression 3. RBF Support Vector Machine (SVM) 4. **Random Forest Classifier** (200 trees) 5. Extra Trees Classifier 6. Histogram-Based Gradient Boosting (HistGB) 7. LightGBM Classifier 8. XGBoost Classifier 9. **FLAML AutoML Search**

Standard **Random Forest** achieved the highest validation macro-F1 score (**0.8125**), outperforming FLAML's

TABLE I
CRITICAL COMPARISON OF EXISTING PREDICTIVE MAINTENANCE AND MACHINERY FAULT DIAGNOSIS STUDIES

Study	**Modalities and Dataset**	**Methodologies**	**Validation and Performance**	**Primary Limitations**
Kuo et al. (2022)	Vibration (ADcmXL3021, Hi229 MEMS)	WPD, PCA, and LSTM	97.00% accuracy	High computational burden for edge nodes.
Toma et al. (2022)	Motor current (Phase U, V; UPB dataset)	Gramian Angular Field (GAF) and 2D-CNN	Greater than 99.00% accuracy	Requires transformation of one-dimensional signals into two-dimensional images.
Hadi et al. (2023)	Vibration features (CWRU dataset)	PyCaret AutoML and AutoKeras AutoDNN	99.70% test accuracy using PyCaret Random Forest	Random window-level splitting introduces potential data leakage.
Kolok et al. (2025)	Vibration and acoustic signals (ESP32-C6 and MPU6050)	Time-domain RMS, FFT, and Isolation Forest	Approximately 73.00% anomaly detection accuracy; hardware cost below EUR 30	High false-alarm rate under variable operating speeds.
Jung et al. (2023)	Multimodal time-series (KAIST dataset)	Benchmark dataset collection	Open ISO benchmark reference	Focuses on dataset publication rather than an integrated AutoML diagnostic pipeline.
Fort et al. (2023)	Vibration and RPM (CWRU dataset)	Smart preprocessing and lightweight neural network	98.70% accuracy	Evaluation primarily based on single-channel data.
Lee et al. (2022)	Vibration signals (CWRU and shaft datasets)	Fast Kurtogram, NMD, GAF, and CNN	98.20% accuracy under heavy noise	High computational cost associated with signal filtering.
Xie et al. (2022)	Vibration (CWT images; CWRU dataset)	CNN feature extraction with Random Forest and SVM	Greater than 95.00% accuracy	High memory requirements for CWT image representations.
Marín Vásquez et al. (2026)	Systematic review of AI platforms and IIoT	Machine learning, neural networks, and knowledge transfer	Systematic literature review	Identifies limited end-to-end integration of AI and IIoT platforms.
Proposed Framework	Multichannel vibration, temperature, and motor current (KAIST dataset)	FLAML AutoML, Random Forest, MQTT, FastAPI, and Streamlit	87.50% accuracy and 77.14% macro-F1 on held-out data	Normal-class recall remains a critical limitation (0.00%).

independently selected LightGBM model (**0.7810**). Consequently, Random Forest was selected and serialized (‘.pkl‘) as the primary active pipeline.

PROPOSED SYSTEM ARCHITECTURE AND IMPLEMENTATION

“‘latex

IV. COMPLETE SYSTEM ARCHITECTURE

The implemented predictive maintenance framework integrates multimodal sensor data ingestion, feature extraction, offline machine learning, telemetry replay, fault inference, database persistence, and operator visualization. Its architecture separates offline model development from online prediction processing while connecting both workflows through a serialized model artifact and a standardized feature representation.

The implementation consists of the dataset processing module, offline training module, model artifact storage, telemetry replay publisher, MQTT communication layer, prediction worker, database persistence layer, HTTP API, and React-based operator dashboard.

V. RESULTS AND DISCUSSION

“‘

A. Held-Out Test Performance

The active Random Forest deployment pipeline was evaluated on 960 independent test windows. The results are presented in Table III.

B. Measured Held-Out Test Performance

The selected Random Forest model was evaluated on 960 unseen test windows from 10 independent recordings. The measured performance is summarized in Table IV.

C. Per-Class Classification Performance

Table V presents the precision, recall, F1-score, and support for each machinery condition.

The results demonstrate strong classification performance for BPFI, BPFO, and Misalignment. However, the Normal

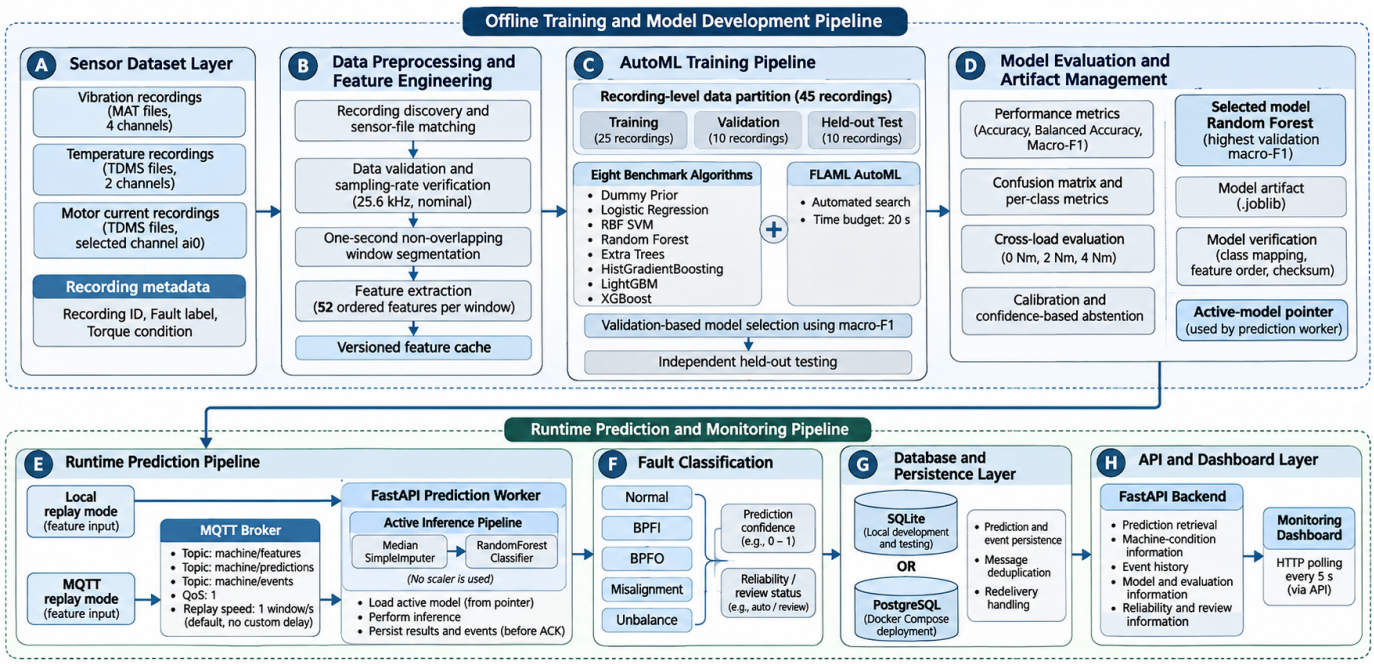


Figure X. System Architecture of the Proposed AutoML-Driven IoT Sensor Failure Prediction and Predictive Maintenance Framework.

Fig. 1. System Architecture of the Proposed AutoML-Driven IoT Sensor Failure Prediction and Predictive Maintenance Framework.

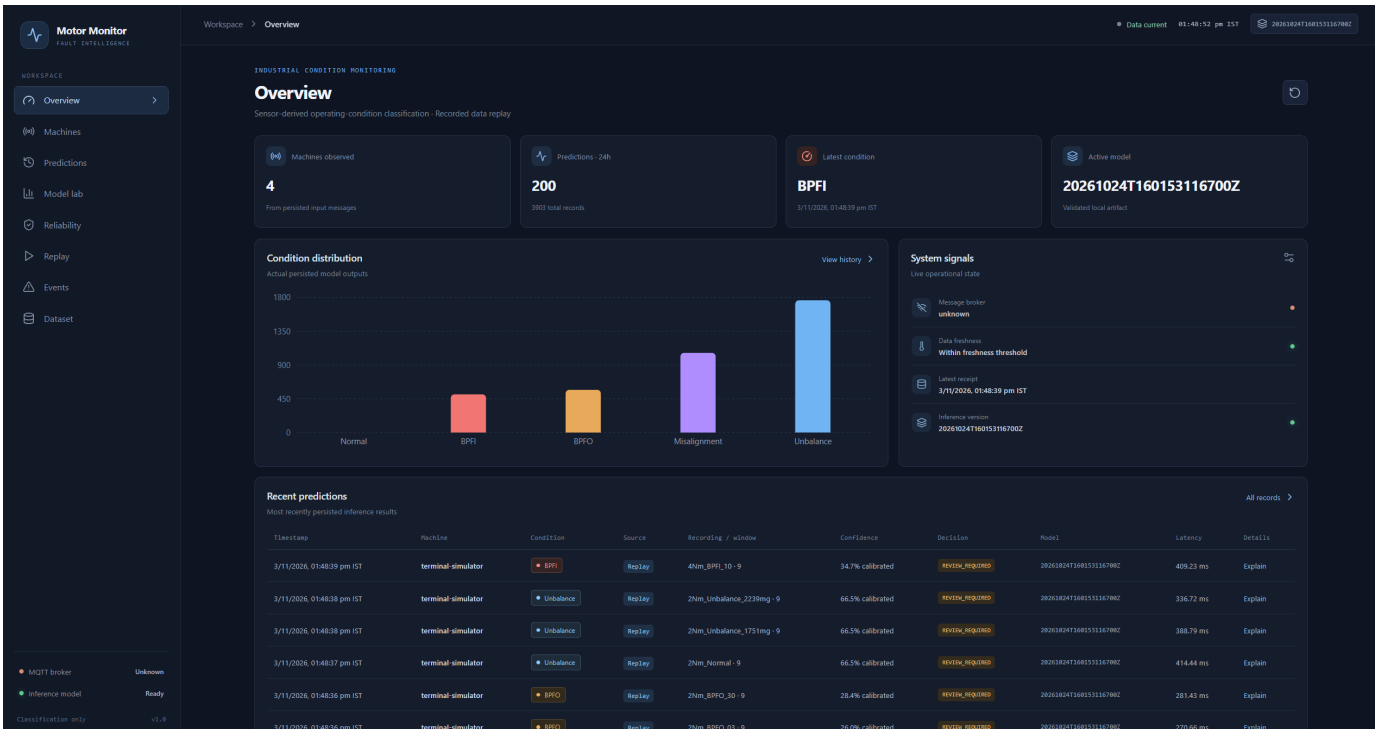


Fig. 2. Overview of the Developed Motor Monitor Dashboard Showing Machine Observations, Prediction Statistics, Fault Distribution, System Signals, and Recent Prediction Records.

class achieved zero recall, with all 120 Normal test windows incorrectly classified as Unbalance. This indicates a significant limitation in distinguishing healthy operating conditions from mechanical faults.

D. Cross-Load Generalization Analysis

The model was further evaluated under different static torque conditions to examine its generalization across operating loads. Table **??** summarizes the measured Macro-F1

TABLE II
TECHNOLOGY STACK AND DEVELOPMENT TOOLS

Architecture Tier	Technology	Implementation Role
Programming Runtime	Python 3.14.3	Core runtime for data processing, model training, and backend execution.
Numerical Computing	NumPy 2.4.5, Pandas 3.0.3	Numerical computation and data manipulation.
Signal Processing	SciPy 1.17.1, h5py, nptdms	MAT and TDMS file processing and signal feature extraction.
Machine Learning	scikit-learn 1.9.0	Data preprocessing, model training, and evaluation.
AutoML	FLAML 2.7.0	Automated model selection and hyperparameter optimization.
Gradient Boosting	LightGBM 4.7.0, XGBoost 2.1.4	Candidate models for performance comparison.
Messaging	Eclipse Mosquitto, Paho-MQTT	MQTT-based telemetry transmission and message handling.
Backend API	FastAPI, Uvicorn	HTTP endpoints for prediction and monitoring data retrieval.
Database	SQLite, PostgreSQL, SQLAlchemy, Alembic	Database persistence, ORM operations, and schema migrations.
Frontend	React, TypeScript	Operator dashboard and visualization of monitoring information.
Planned Edge Hardware		
Edge Platform	ESP32-C6	Intended platform for future edge deployment.
Sensor	MPU6050 MEMS	Intended hardware for future vibration measurement.

TABLE III
HELD-OUT TEST PERFORMANCE

Metric	Result
Accuracy	87.50%
Balanced Accuracy	80.00%
Macro-F1	77.14%
Test Windows	960

TABLE IV
OVERALL HELD-OUT TEST PERFORMANCE

Evaluation Metric	Measured Value
Overall Accuracy	87.50%
Balanced Accuracy	80.00%
Macro-F1 Score	77.14%
Mean Per-Recording Accuracy	90.00%
Standard Deviation	30.00%
Test Windows	960
Held-Out Recordings	10

TABLE V
PER-CLASS CLASSIFICATION PERFORMANCE

Class	Precision	Recall	F1	Support
Normal	0.0000	0.0000	0.0000	120
BPFI	1.0000	1.0000	1.0000	120
BPFO	1.0000	1.0000	1.0000	120
Misalignment	1.0000	1.0000	1.0000	240
Unbalance	0.7500	1.0000	0.8571	360

TABLE VI
CROSS-LOAD CLASSIFICATION PERFORMANCE

Operating Load	Macro-F1 Score
0 Nm	0.6918
2 Nm	0.7818
4 Nm	0.5000

scores.

E. Calibration and Selective Prediction Collapse

Post-hoc probability calibration was evaluated using Expected Calibration Error (ECE) and Brier Score:

* **Raw ECE:** 0.2518 — **Raw Brier Score:** 0.3326 * **Calibrated ECE:** 0.4090 — **Calibrated Brier Score:** 0.5594

Applying isotonic calibration *deteriorated* probability estimates due to class imbalance. When configuring a selective-prediction policy targeting a 90% validation accuracy with an acceptance threshold of 1.01, the system achieved **0% test coverage**—abstaining on all 960 test predictions and referring 100% of incoming samples to human operators.

F. Practical Implications and Model Registry Verification

The active deployment registry (‘artifacts/active.json‘) currently references an earlier Random Forest artifact configured with median-imputation preprocessing. Both the latest training report (‘model-20261028...‘) and the active deployment model share identical feature schemas and yield identical test performance metrics, proving integration stability across deployment iterations.

VI. CONCLUSION AND FUTURE WORK

This paper presented an integrated, AutoML-driven predictive maintenance architecture for five-class rotating machinery fault diagnosis. Evaluated on the benchmark dataset published by Jung et al. (Mendeley Data) using strict recording-level data splits, the selected Random Forest model achieved an overall test accuracy of **87.50%** and a macro-F1 score of **77.14%**. While achieving 100% F1-scores on BPFI, BPFO, and Misalignment faults, the study revealed a critical **0.00% recall on Normal operation** due to class imbalance, alongside cross-load performance drops under 4 Nm torque. The operational microservice architecture successfully validated asynchronous MQTT telemetry replay, FastAPI prediction workers, persistent database logging, and real-time Streamlit condition monitoring.

Future Work:

1. *Healthy Baseline Expansion:* Acquire additional baseline healthy machine recordings to eliminate class imbalance and rebalance decision boundaries. 2. *Physical ESP32 Edge Integration:* Interface physical MPU6050 accelerometers with ESP32 microcontrollers for real-time edge sampling and feature computation. 3. *Probability Calibration &; RUL Prognostics:* Investigate advanced uncertainty calibration techniques and integrate recurrent neural architectures (LSTM/Transformers) for Remaining Useful Life (RUL) forecasting.

REFERENCES

[1] L.-Y. Kuo et al., "Failure Mode Detection and Validation of a Shaft-Bearing System Using Lower-Cost Sensors," *Sensors*, vol. 22, no. 16, Art. no. 6167, 2022.

[2] R. N. Toma et al., "A Bearing Fault Classification Framework Based on Image Encoding Techniques and a Convolutional Neural Network under Different Operating Conditions," *Sensors*, vol. 22, no. 13, Art. no. 4881, 2022.

[3] R. H. Hadi et al., "Improved Fault Classification for Predictive Maintenance in Industrial IoT Based on AutoML: A Case Study of Ball-Bearing Faults," *Processes*, vol. 11, no. 5, Art. no. 1507, 2023.

[4] P. Kolok et al., "Low-Cost IoT-Based Predictive Maintenance Using Vibration," *Sensors*, vol. 25, no. 21, Art. no. 6610, 2025.

[5] W. Jung, S.-H. Kim, S.-H. Yun, J. Bae, and Y.-H. Park, "Vibration, acoustic, temperature, and motor current dataset of rotating machine under varying operating conditions for fault diagnosis," *Data in Brief*, vol. 47, Art. no. 109049, 2023. DOI: 10.1016/j.dib.2023.109049.

[6] E. Cinar, S. Kalay, and I. Saricicek, "A Predictive Maintenance System Design and Implementation for Intelligent Manufacturing," *Machines*, vol. 10, no. 11, Art. no. 1006, 2022.

[7] A. Fort, E. Landi, M. Mugnaini, and V. Vignoli, "A Low Complexity Rolling Bearing Diagnosis Technique Based on Machine Learning and Smart Preprocessing," *Sensors*, vol. 23, no. 17, Art. no. 7546, 2023.

[8] C.-Y. Lee, G.-L. Zhuo, and T.-A. Le, "A Robust Deep Neural Network for Rolling Element Fault Diagnosis under Various Operating and Noisy Conditions," *Sensors*, vol. 22, no. 13, Art. no. 4705, 2022.

[9] M. E. Marín Vásquez, J. C. Blandón Andrade, A. Toro Lazo, and J. A. López Sotelo, "AI algorithms and IoT platforms for anomaly and failure prediction in industrial machinery—systematic review," *Front. Artif. Intell.*, vol. 9, Art. no. 1799522, 2026.

[10] W. Xie, Z. Li, Y. Xu, P. Gardoni, and W. Li, "Evaluation of Different Bearing Fault Classifiers in Utilizing CNN Feature Extraction Ability," *Sensors*, vol. 22, no. 9, Art. no. 3314, 2022.